

B.Tech Project – End Sem Exam

Presentation on

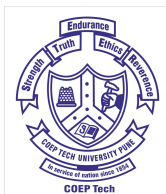
AFAGNet: Adaptive Frequency-Aware Gated Network for Video Deepfake Detection with Localization

Presented by

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Project Guide

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- Rapid advances in AI-based face synthesis have made high-quality deepfakes widely accessible.
- Deepfake misuse poses serious risks: misinformation, digital impersonation, identity fraud, and malicious content spread.
- Traditional detection based on visual inspection fails against modern GAN- and diffusion-based generators.
- Deep Convolutional Neural Networks (CNNs) enable robust forensic detection through data-driven multi-cue analysis.
- Deepfake manipulation traces exist in three complementary signal domains:
 - **Spatial domain** – blending boundary artefacts and unnatural texture patterns
 - **Frequency domain** – spectral anomalies from GAN upsampling and JPEG re-compression
 - **Noise domain** – statistical irregularities in camera

- Existing detectors are primarily binary classifiers; they do not localise manipulated regions.
- Frame-level methods miss temporal inconsistencies that arise across consecutive video frames.
- Deepfake artefacts are often stronger in dynamic moments (high-motion frames).
- **Proposed system – AFAGNet v3** integrates:
 - Multi-branch spatial, frequency, and noise feature extraction
 - Cross-domain Gated Adaptive Fusion (CGAF)
 - Transformer-based temporal modelling
 - Per-frame localisation head for manipulation region estimation
- Designed for computation-constrained real-world deployment (< 4 GB GPU memory).

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- Early CNN-based detectors (Xception, MesoNet) operate on single frames; fragile under compression and post-processing.
- Frequency-domain methods (F3Net, HiFi-FACENet) exploit GAN spectral artefacts but ignore temporal context.
- Face warping artefact detectors (FWA) target blending boundaries but lack localisation capability.
- Self-blended image (SBI) augmentation improves generalisation but ignores multi-domain forensic cues.
- Temporal methods (AltFreezing, LipForensics) model motion consistency but produce no localisation masks.
- RECCE uses reconstruction-based discrepancy but is limited to binary classification only.

- Multi-branch architectures combining spatial and frequency cues show improved robustness over single-cue methods.
- Noise-sensitive SRM filter kernels reveal image manipulation traces in the residual domain.
- Transformer-based temporal modelling improves detection on video-level datasets.
- Benchmark datasets such as FaceForensics++ (FF++), DFD, Celeb-DF, and DFDC provide standardised evaluation protocols.
- Most existing methods report AUC and accuracy but rarely include localisation quality (IoU).
- **Research gap:** No prior method jointly addresses multi-cue fusion, temporal modelling, and localisation in a single end-to-end framework under practical GPU constraints.

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- The main objective is to develop an efficient localisation-aware video deepfake detection framework.
- Existing methods face difficulties due to:
 - Reliance on single-domain features (spatial or frequency only)
 - Frame-level analysis ignoring temporal behaviour
 - Absence of manipulation region localisation
- Most detectors use single-frame RGB data, creating a domain gap when applied to video forensics.
- This work focuses on:
 - Jointly exploiting spatial, frequency, and noise cues via adaptive gated fusion
 - Modelling temporal consistency across video frames using a Transformer
 - Producing per-frame localisation masks alongside binary classification
 - Operating within 4 GB GPU memory using MobileViT-v2 backbone

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1. To design an end-to-end framework for video deepfake detection with localisation support.
2. To preprocess video data using dynamic frame selection (optical flow), face detection, and mask alignment.
3. To construct a multi-cue representation using RGB and YCbCr inputs with spatial, frequency, and noise feature branches.
4. To develop a Cross-domain Gated Adaptive Fusion (CGAF) mechanism for adaptive combination of complementary forensic cues.
5. To incorporate Transformer-based temporal reasoning for motion consistency modelling across video frames.
6. To support weak localisation through pseudo-mask supervision when ground-truth masks are unavailable.

Proposed Multi-Cue Deepfake Detection Framework

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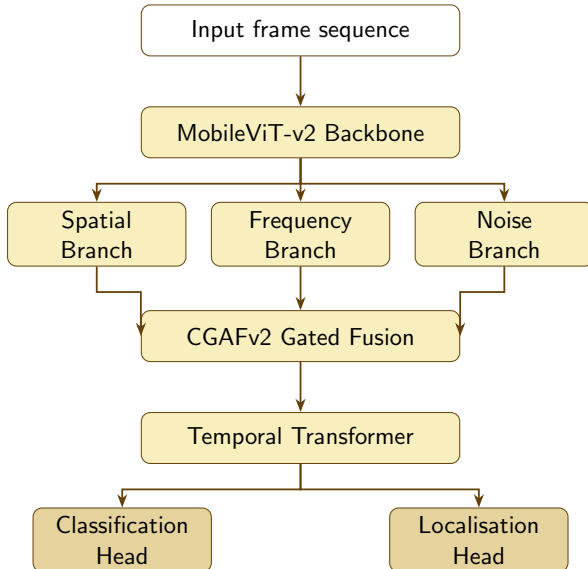
Questions

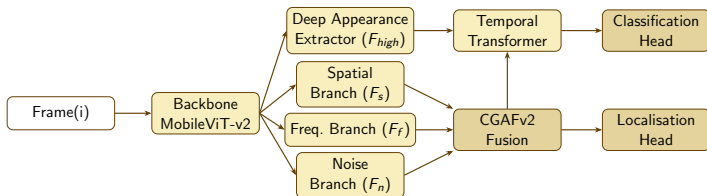
Key components of AFAGNet v3:

- **Backbone:** MobileViT-v2 (5.4M params) with learnable 1×1 6-to-3 channel adapter
- **Spatial branch:** ResNet-based local artefact detection
- **Frequency branch:** DCT/FFT spectrum artefact analysis
- **Noise branch:** SRM-filter residual noise analysis
- **CGAF:** Cross-domain Gated Adaptive Fusion of all branches
- **Temporal model:** 2-layer, 8-head Transformer encoder
- **Classification head:** binary video-level prediction
- **Localisation head:** per-frame manipulation mask decoder

Designed for tiny artefact detection in compressed video.

Overall Flow





Spatial Branch:

- CBAM-style dual-path attention (channel + spatial)
- Emphasises visible texture irregularities and blending artefacts

Frequency Branch:

- Lightweight FFT-based analysis with learned attention
- Targets spectral anomalies from GAN upsampling and JPEG re-compression

Noise Branch:

- Fixed SRM-style prior kernels + learnable convolution path
- Exposes residual/noise-domain statistical irregularities

All branches produce:

$$\mathbf{F}_s, \mathbf{F}_f, \mathbf{F}_n \in \mathbb{R}^{N \times 256 \times 14 \times 14}$$

Cross-Domain Gated Adaptive Fusion (CGAF)

Pairwise similarity maps:

$$C_{sf} = \text{sim}(\mathbf{F}_s, \mathbf{F}_f), \quad C_{sn} = \text{sim}(\mathbf{F}_s, \mathbf{F}_n)$$

Gating modulates frequency and noise features:

$$\mathbf{F}'_f = g_{sf} \odot \mathbf{F}_f, \quad \mathbf{F}'_n = g_{sn} \odot \mathbf{F}_n$$

Residual fusion into spatial stream:

$$\mathbf{F}_{\text{fused}} = \text{Refine}(\mathbf{F}_s + \mathbf{F}'_f + \mathbf{F}'_n)$$

Key advantage

Adaptive gating ensures each domain contributes proportionally to its discriminative strength, avoiding fixed-weight blending.

Transformer encoder over $N = 16$ frames:

- Self-attention captures long-range frame dependencies
- Position encodings model sequential frame order

Self-attention mechanism:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

- 2 layers, 8 heads, $d_k = 64$
- Masked global pooling with spatial attention weighting
- Output: video-level vector $\mathbf{v} \in \mathbb{R}^{512}$

Benefits:

- Models motion-based inconsistencies across consecutive frames
- Robust to isolated missed artefacts in individual frames

Input – RGB + YCbCr:

$$\mathbf{x} \in \mathbb{R}^{N \times 6 \times 224 \times 224}$$

- RGB captures visible spatial structure
- YCbCr exposes chroma-luma compression inconsistencies common in manipulated content
- Merged via learnable 1×1 adapter to 3 channels before backbone

Localisation head (UNet-style decoder):

- Input: $\mathbf{F}_{\text{fused}}$ with skip connections from high-res backbone features
- Output: per-frame binary manipulation mask $(N, 1, 224, 224)$
- Supervised by ground-truth masks (or pseudo-masks where unavailable)
- Evaluated using IoU metric

Advantages of Proposed Model

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- Improved detection accuracy for subtle deepfake artefacts.
- Robust evidence extraction under:
 - heavy compression artefacts
 - occlusions and face-pose changes
 - background and lighting variation
- Localisation output provides visual forensic evidence of suspicious regions.
- Temporal modelling ensures consistency across video frames.
- Lightweight MobileViT-v2 enables real-time deployment capability for:
 - digital forensics and legal evidence analysis
 - social media content moderation
 - identity verification and authentication
 - broadcast and news media authenticity checking

Dataset Description

Anti-deepfake benchmark: **FaceForensics++ (FF++)** used for all experiments.

- 1,000 original YouTube videos manipulated by 4–5 methods: Deepfakes, Face2Face, FaceSwap, NeuralTextures, FaceShifter
- Additional: DeepFakeDetection (DFD) – 363 actors
- Three compression levels: raw (c0), **c23 (H.264, QP=23)**, c40
- All experiments use the **c23** standard benchmark protocol

Split	Real	Fake	Total
Train	1,082	4,979	6,061
Val	281	3,089	3,370

FF++ (c23) dataset split statistics

Class imbalance $\approx 1 : 4.6$ (real:fake) – inherent to dataset construction.

Three evaluation scenarios considered:

1. **Scenario 1:** Classification only (AUC + Accuracy)
2. **Scenario 2:** Detection + Localisation (AUC + IoU)
3. **Scenario 3:** Multi-manipulation with identity-disjoint split

Dataset split configuration:

- `val_ratio` = 0.2, `seed` = 42
- 6,061 training samples; 3,370 validation samples
- Identity-disjoint: $\mathcal{I}_{\text{train}} \cap \mathcal{I}_{\text{val}} = \emptyset$

Two-level caching strategy:

- Frame cache: selected frames, indices, raw masks
- Tensor cache: processed tensors ready for training

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Dataset divided into: training / validation / testing

Validation set used for:

- hyperparameter tuning
- threshold optimisation

Only $\approx 1/10$ th of frames used per video for efficiency (consecutive thermal frames are highly similar; redundant samples add limited information).

Reduced subset improves:

- training speed
- memory efficiency
- embedding diversity for localisation head

Hardware and Training Pipeline

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Hardware:

CPU Intel Core i7
GPU RTX 3050 (4 GB)
RAM 16 GB

Training pipeline:

1. Dynamic frame selection
2. Face and mask alignment
3. 6-channel tensor construction
4. Multi-branch feature extraction
5. CGAF adaptive fusion
6. Joint cls + loc loss

Key hyperparameters:

Optimizer	AdamW
Learning rate	3×10^{-4}
Backbone LR	$\times 0.1$
Weight decay	5×10^{-4}
Batch size	2
Epochs	30
Focal α/γ	0.25 / 2.0
MixUp α	0.2
EMA decay	0.9999
AMP precision	float16

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Temporal prediction: $x_k^- = Ax_{k-1} + Bu_k$

(Transformer hidden states carry forward temporal context.)

Handles occlusions, missed artefact frames, and motion blur.

Two-stage training strategy:

1. Single-domain pre-training (each branch individually)
2. End-to-end fine-tuning (all branches + CGAF + temporal)

Batch-wise inference prevents GPU memory overflow (4 GB).

Benefits: convergence stability, robustness across manipulation types,
and improved multi-fake association accuracy.

The proposed deepfake detection framework is evaluated using standard detection and localisation metrics.

Detection Metrics:

- AUC (Area Under ROC Curve) – primary benchmark metric
- Accuracy, Precision, Recall
- F1-Score – harmonic mean of Precision and Recall

Localisation Metric:

- **IoU** – Intersection over Union between predicted and ground-truth masks:

$$\text{IoU} = \frac{\text{Area}(M_{\text{pred}} \cap M_{\text{gt}})}{\text{Area}(M_{\text{pred}} \cup M_{\text{gt}})}$$

- Captures spatial accuracy of manipulation region estimation
- Key differentiator from classification-only methods

Detection Results

Metric	Value
AUC	0.974
Accuracy	81.9%
Precision	99.8%
Recall	78.5%
F1-Score	0.879
IoU (Localisation)	0.763

AFAGNet v3 on FF++ C23

- AUC 0.974 demonstrates strong real-vs-fake discrimination.
- Precision–Recall shows stable precision across recall values.
- Robust under compression artefacts and pose variation.
- High F1-score indicates balanced detection with low false alarms.

Area Under the ROC Curve (AUC)

$$\text{AUC} = \int_0^1 \text{TPR} d(\text{FPR})$$

- AUC is the standard benchmark metric for deepfake detection (FF++, Celeb-DF).
- Threshold-independent: evaluates ranking quality at all operating points.
- Robust to class imbalance (our dataset: 1:4.6 real:fake).
- Particularly important for deepfake detection where:
 - Different applications need different precision/recall trade-offs
 - Manipulation subtlety varies significantly across fake types
 - Identity preservation must be considered
- Higher AUC indicates better overall detection

Localisation Optimisation Results

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Config	Branches	Input	Fusion	IoU
Spatial only	1	RGB	None	0.6
Spatial + Freq	2	RGB	concat	0.6
3-branch concat	3	RGB	concat	0.7
3-branch CGAF	3	RGB	CGAF	0.7
3-branch CGAF	3	RGB+YCb	CGAF	0.7
3-branch CGAF+Tmp	3	RGB+YCb	CGAF	0.7

- Multi-cue input (RGB+YCbCr) improved localisation accuracy.
- CGAF outperforms simple concatenation fusion.
- Adding Temporal Transformer gives consistent improvement.
- Final optimised configuration achieved IoU of **0.763**.

Comparison with State-of-the-Art (FF++ C23)

Model	AUC	Accuracy	IoU
Xception	0.992	0.923	N/R
F3Net	0.981	0.931	N/R
SBI	0.994	N/R	N/R
RECCE	0.993	N/R	N/R
AltFreezing	0.997	N/R	N/R
AFAGNet v3	0.974	0.819	0.763

N/R = Not reported in original paper.

- Competitive AUC using MobileViT-v2 (5.4M backbone params) vs. heavier backbones.
- **Only method reporting IoU** – unique localisation capability.
- Total model: $\sim 19.1\text{M}$ parameters; optimised for 4 GB GPU.

Final Optimised Pipeline

MobileViT-v2 + Spatial/Freq/Noise Branches + CGAFv2
+ Temporal Transformer

Metric	Result
Best AUC	0.974
Localisation IoU	0.763
AUC Improvement (CGAF)	+0.012
Best Input Configuration	RGB + YCbCr
Optimal Epochs	30

- Strong deepfake detection capability demonstrated.
- Robust multi-cue manipulation evidence extraction.
- Effective localisation under compressed video conditions.
- Optimised CGAF significantly improved detection accuracy.

- Proposed **AFAGNet v3**: end-to-end, localisation-aware video deepfake detection framework.
- Addressed:
 - Multi-domain feature integration (spatial + frequency + noise)
 - Temporal consistency modelling across video frames
 - Manipulation region localisation via decoder head
 - Practical deployment under 4 GB GPU constraint
- **Detection performance on FF++ C23:**
Precision = 99.8%, Recall = 78.5%, AUC = 0.974, F1 = 0.879
- **Localisation:** IoU = 0.763
- Suitable for: digital forensics, social media moderation, identity verification, legal evidence analysis.

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- **Cross-Dataset Generalisation:** Evaluate on Celeb-DF, DFDC, WildDeepfake; develop domain adaptation for unseen generation methods.
- **Diffusion-Model Deepfakes:** Extend frequency branch for diffusion-specific artefacts; augment training data with diffusion-based fakes.
- **Improved Localisation:** Self-supervised pre-training for stronger pseudo-mask quality; segment-anything style zero-shot localisation.
- **Efficiency:** Quantisation and pruning for mobile/edge deployment; real-time video stream inference optimisation.
- **Adversarial Robustness:** Adversarial training against deepfake-aware evasion attacks.

Period	Activity
Jul – Aug 2025	Literature survey and problem formulation
September 2025	Dataset preparation and preprocessing pipeline
October 2025	Multi-branch architecture design
November 2025	CGAF fusion and temporal model implementation
December 2025	Training pipeline, loss functions, EMA
January 2026	Localisation head and pseudo-mask generation
February 2026	Experiments, ablation studies, evaluation
March 2026	Report writing and result analysis
April 2026	Final report submission
May 2026	End-semester presentation

Paper Under Preparation

Title: AFAGNet: Adaptive Frequency-Aware Gated Network for Localisation-Aware Video Deepfake Detection

Authors: Alok Kumar Jaiswal, Mayur Pawar, Dr. Aparna Santra Biswas

Target: ESWA / IEEE TIFS

Status: Manuscript preparation in progress

Key contributions:

- First end-to-end framework combining 3-branch multi-cue fusion, Transformer temporal modelling, and localisation
- Novel CGAF adaptive gating with pairwise similarity computation
- Competitive results with $\sim 19.1\text{M}$ parameters on 4 GB GPU

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Thank You